

Section 6

Redshift Security

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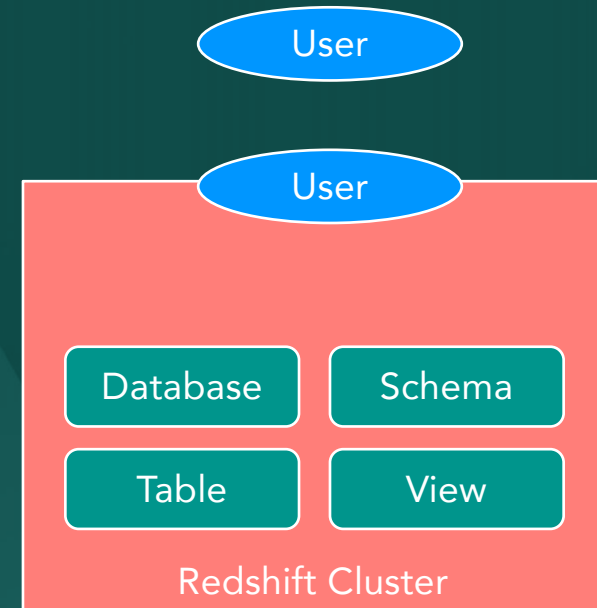


Agenda

- Users & Groups
- Permissions & Roles
- RBAC
- Row Level Security (RLS)
- Column Level Security (CLS)
- Dynamic Data Masking
- Encryption
- Audit Logging
- Lake Formation

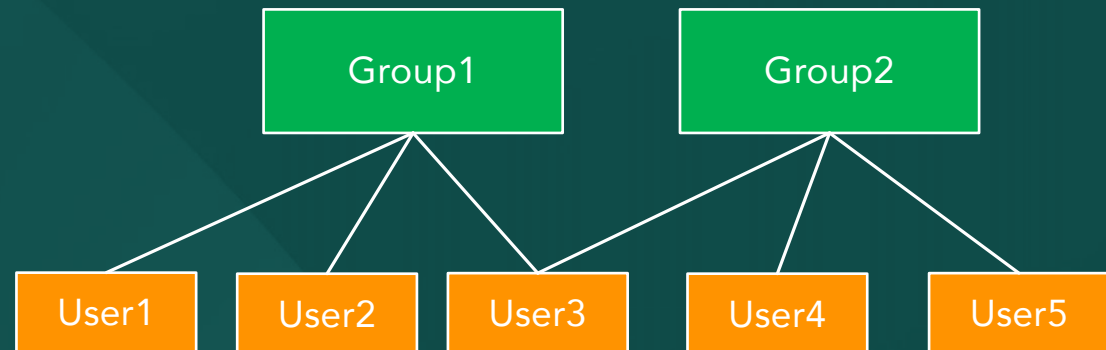
Authentication and Authorization

- Authentication is getting into the system.
- Authorization is accessing the resources in the system.
- Users
 - Superuser
 - Normal users
- Groups – Collection of users.
- Roles – Collection of permissions.
- Permissions – What users and groups can do.



Users & Groups

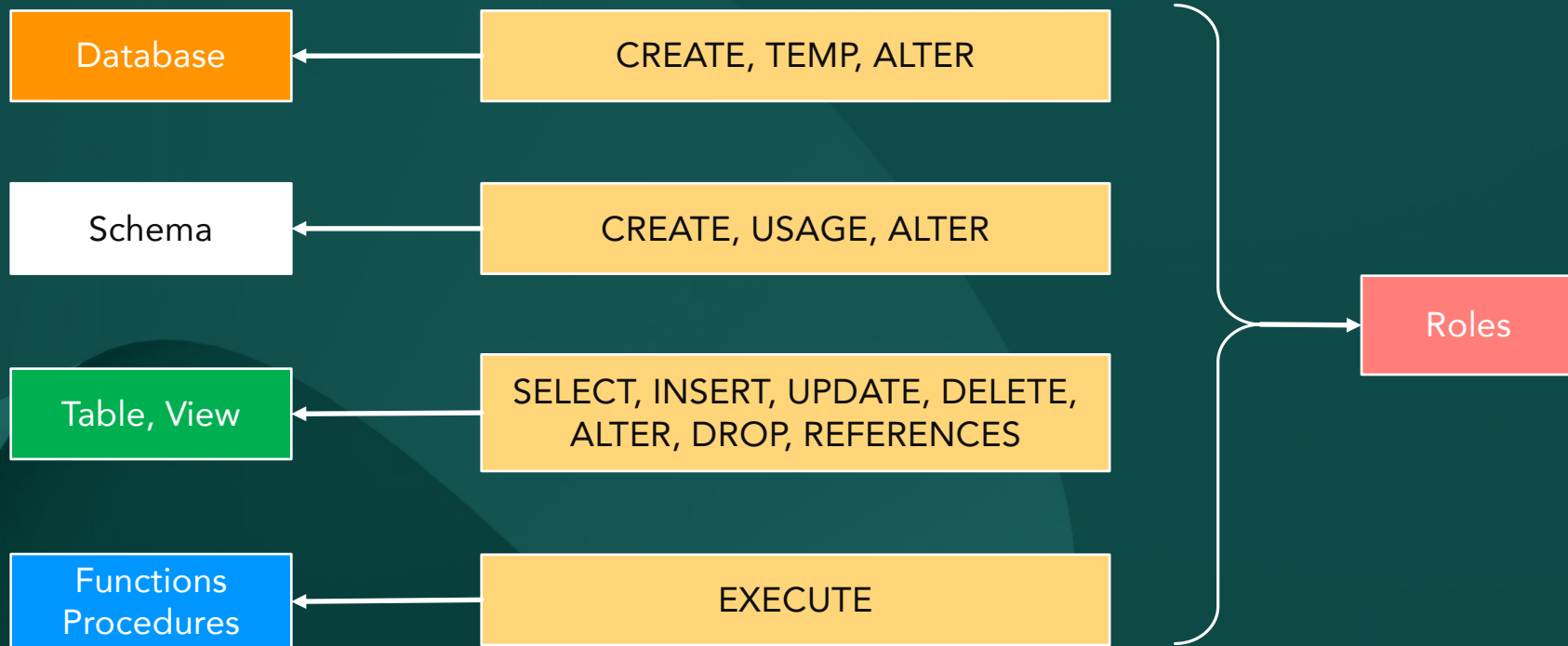
- Users can enter system and access system resources.
- Users can be put in groups.
- Groups make authorization management easier.
- svv_user_info, pg_user
- pg_group
- Additional super users can be created too.
 - CREATE USER admin
CREATEUSER PASSWORD '...';



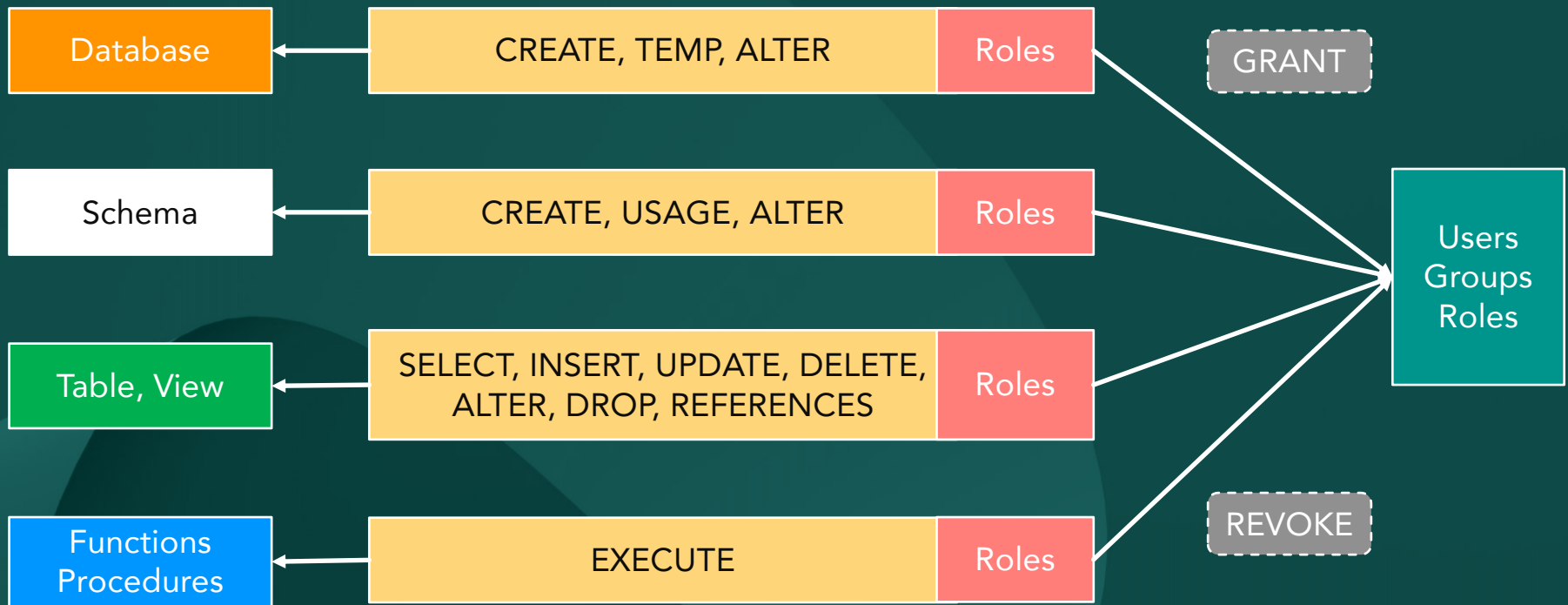
```
CREATE USER <user_name>  
WITH PASSWORD 'password'  
IN GROUP <group_name>  
CONNECTION LIMIT <n> | UNLIMITED  
SESSION TIMEOUT <seconds>
```

```
CREATE GROUP <group_name>  
WITH USER <user1>, <user2>
```

Permissions

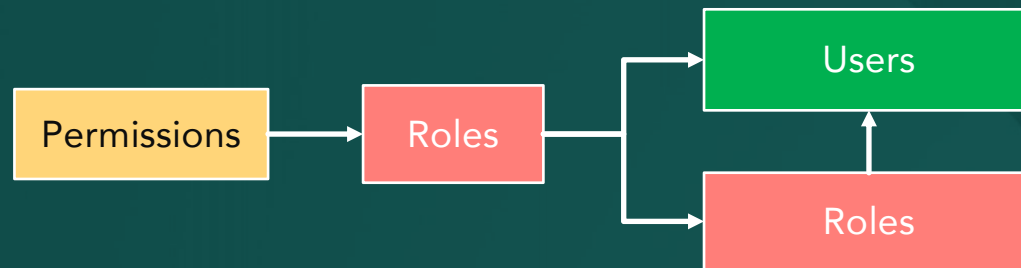


Assignment

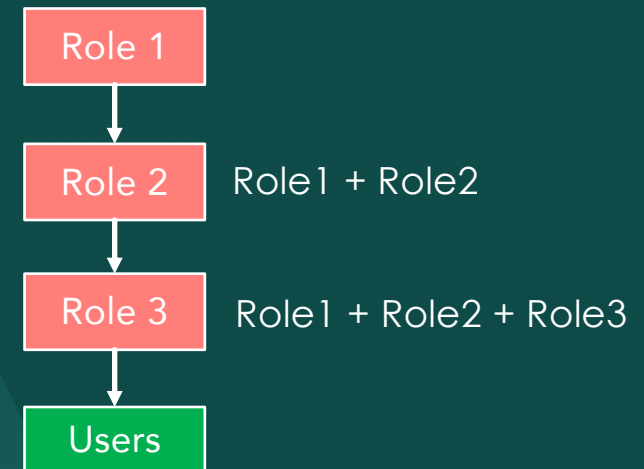


Roles can only be assigned to Users

Role Based Access Control (RBAC)



- RBAC is Providing access using Roles.
- Why Roles and not Groups.
 - Gives more flexibility on permission management.
- GRANT and REVOKE statements.
- SVV_ROLES.



```
CREATE ROLE <role_name>  
ALTER ROLE ...
```

Redshift Defined Roles

- `sys:monitor` – Access catalog and system tables (SVV, SYS, STL, STV, SVCS, SVL, PG_*).
- `sys:operator` – Access catalog & system tables, execute ANALYZE and VACUUM commands, cancel SQL queries.
- `sys:dba` – Create schemas, tables, views, drop schemas, drop tables, truncate tables. Create and drop stored procedures, functions. All permissions from `sys:operator` role.
- `sys:superuser` – All permissions.
- `sys:secadmin` – Create, drop, alter users and roles.

Hands-on

- Create users – u_rpt, u_dsc, u_mle, u_dana, u_adm, u_dsh.
- Create groups – dashboard, reporting, dba, application.
- Create roles – role_dsh, role_ana, role_admin.
- Lets check the access of all users. Will login using RSQL.
- Grant read access of orders, order_items tables to u_rpt and u_dsc users.
- Grant read and write of all tables to dba group.
- Grant read role of all tables to role_dsh and role_ana roles.
- Grant read and write of all tables to role_admin role. Attach this role to u_adm, u_dana users.

Row Level Security (RLS)

- Give permission to subset of data.
- Users and roles can be given access to specific records/rows.
- RLS policy.
- System tables/views
 - SVW_ROLES
 - SVW_RLS_POLICY
 - SVW_RLS_ATTACHED_POLICY

```
CREATE RLS POLICY <policy_name>  
WITH (<column> data_type)  
USING (<expression similar to WHERE clause>)
```



Column Level Security (CLS)

- Permission to subset of columns.

```
GRANT SELECT | UPDATE <column-name>  
ON TABLE <table-name>  
TO <user> | ROLE <role-name> | GROUP <group-name>
```

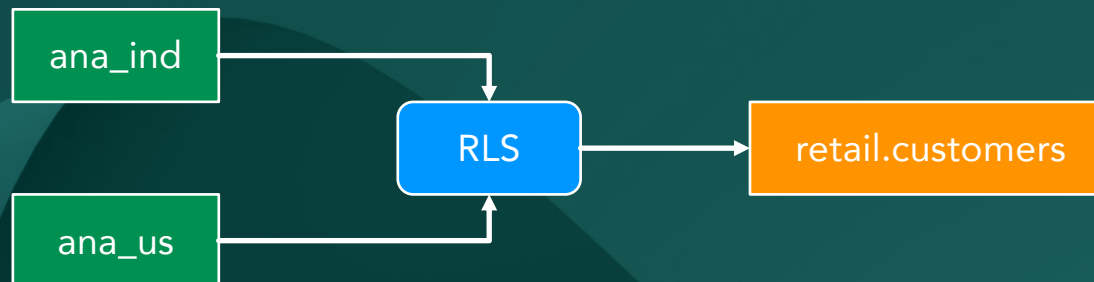
Hands-on

- User u_mle should have access to products that are priced more than 100,000.



- Lets join orders table

- Create ana_ind and ana_us roles. ana_ind should have access to orders that have emanated from India, ana_us should have access to orders that are from USA.

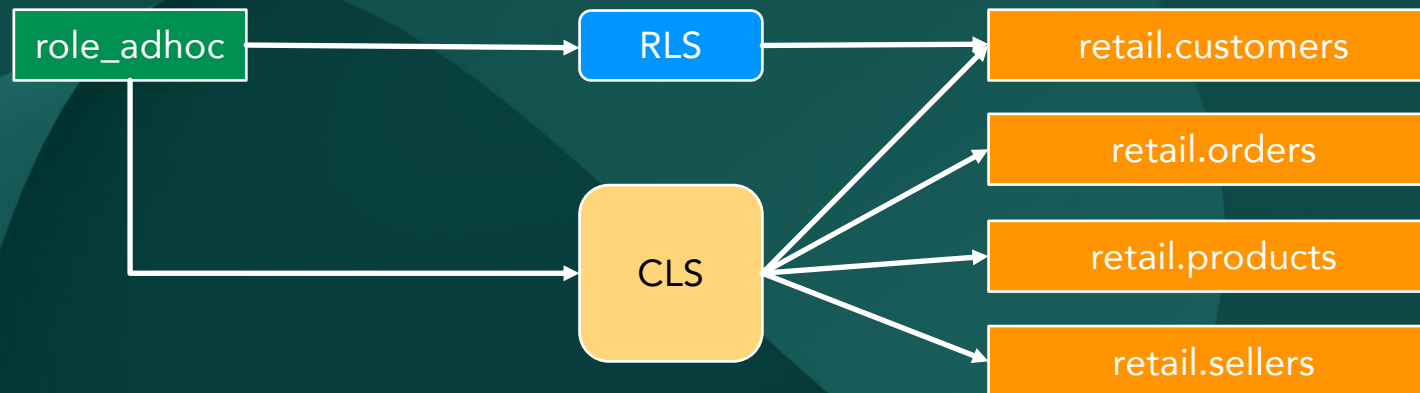


- Lets join orders, customers and products table

- DBAs should not have access to read sensitive data like customer_id, seller_id, product_id, price, product_review.

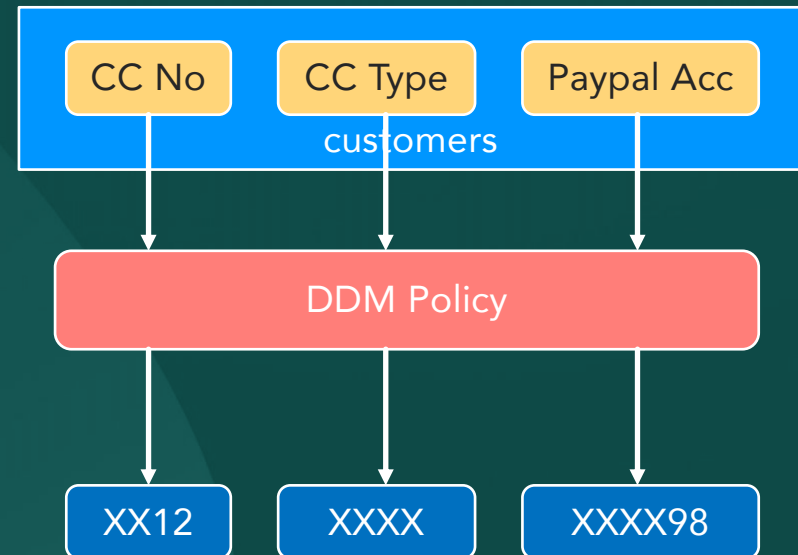
Hands-on

- Role `role_adhoc` should have access to rows belonging to country USA. Also, this role (with user `u_adh`) should be able to see only the following mentioned columns:
 - Orders – `order_id`, `order_date`.
 - Products – `product_id`, `product_name`, `product_price`.
 - Sellers – `seller_id`, `seller_active_date`, `seller_business_type`, `seller_company_name`.
 - Customers – `customer_id`, `email`, `phone`.



Dynamic Data Masking (DDM)

- Credit Card No – XXXX XXXX XXXX 1234
- Phone – XXXXXXXX098
- DOB – XX/XX/1993
- DDM is used to MASK sensitive data.
- Can be done with SUPER data type also.
- Conditional masking.
- System views
 - SW_MASKING_POLICY
 - SW_ATTACHED_MASKING_POLICY
 - SYS_APPLIED_MASKING_POLICY_LOG



Dynamic Data Masking (DDM)

- Create policy

```
CREATE MASKING POLICY <policy-name>  
WITH (<input-columns>)  
USING (<expression>);
```

- Attach policy

```
ATTACH MASKING POLICY <policy-name>  
ON <table-name> (column names)  
TO <user> | ROLE <role-name>  
PRIORITY <priority>;
```

Hands-on

- Make sure all roles and users except superuser fetch only masked data for the following information:
 - Orders – payment_method
 - Sellers – seller_business_type and seller_company_name
 - Customers – company, dob, credit_card_no, credit_card_type, paypal_account
 - Products – product_price

Encryption

- Data encryption at rest (Server side encryption).
 - Redshift encrypts while writing.
 - Redshift decrypts while reading.
- Unencrypted cluster → Encrypted cluster
 - Data is migrated from uncr to encr cluster. Cluster is available for READs ONLY.
- Encryption at transit.
 - SSL connections
 - From JDBC, ODBC connections to Redshift cluster.
 - Hardware accelerate SSL for communication between S3 and DynamoDB.
- Network security – Between Redshift and S3 using VPC Endpoint.
- Data tokenization is supported for 3rd party services using AWS Lambda.

Audit Logging

- Logs connections and user activities in the database.
- Stored in S3 or CloudWatch Logs.
- Connection Log – Authentication and database connections.
 - STL_CONNECTION_LOG
 - SYS_CONNECTION_LOG
- User Log
 - SYS_USERLOG
 - STL_USERLOG
- User Activity Log
 - enable_user_activity_logging
- CloudWatch Logs – /aws/redshift/cluster/<cluster>/<log-type>

Hands-on

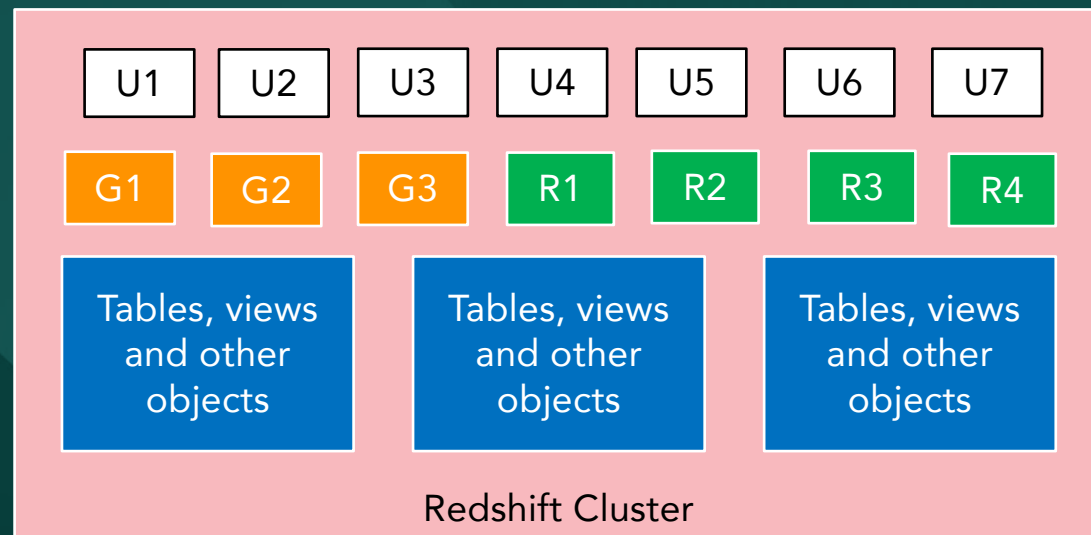
- Check logging in CloudWatch Log Groups

Lake Formation

- Tool to centrally govern and secure data for the following:
 - Redshift Spectrum
 - Athena
- Row level, Column level permissions.

Implement and Usage

- Create cluster.
- Create Groups and Roles as per data governance requirement.
- Add users to groups.
- Attach roles to users.



Authentication with IdP and SSO

- Azure Active Directory
- SSO using IAM Identity Center

Summary

- Authentication & authorization
- Users, Groups and Roles
- Permissions at object level
- Permission assignment
- RBAC
- RLS, CLS, DDM
- Encryption
- Lake Formation

Thank You

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